

Revision 5 Security Audit Remediation Report

(§26)

Assessment Date: September 2026

Scope: Full Stack Platform (Web SSR, Android Mobile Native, Distributed Backend Gateway, PostgreSQL RDS)

Status: ALL 10 FINDINGS FULLY REMEDIATED & VALIDATED

Release Blocker Status: RESOLVED

Executive Summary

This document certifies the comprehensive audit remediation of the 10 security vulnerabilities and operational risks identified in the **NetworkPeer Revision 5 Architecture & Security Review (§26)**. All high, medium, and low severity findings have been addressed in code, verified with automated CI guards, and validated on live production environments.

Audit Findings & Remediation Matrix

Finding ID	Severity	Description	Remediation Status	Verification Method
SEC-01	High	Passwordless Email OTP Rate Limiting & Tampering	REMEDIATED	Negative test: 5-attempt lockout & 60s cooldown
SEC-02	CRITICAL	Server-Side Worker Role Review Queue Gating (Release Blocker)	REMEDIATED	Negative test: HTTP 403 Forbidden verified for unapproved workers
SEC-03	Medium	Proprietary AI Model Name Leakage in Client Bundles	REMEDIATED	Automated CI script: <code>scripts/check-model-leak.sh</code> (0 leaks)

Finding ID	Severity	Description	Remediation Status	Verification Method
SEC-04	Medium	Missing Mandatory Identity Fields (Name / Phone)	REMEDIATED	Schema & form validation: min 2 chars, E.164 phone
SEC-05	High	Client Hydration Outage & Unhandled Root Boundary	REMEDIATED	Portal layout restoration, client-only guards on IndexedDB/WS
SEC-06	High	Unfunded Escrow Job Dispatch Vulnerability	REMEDIATED	State machine enforcement: Jobs require FUNDED status to dispatch
SEC-07	Medium	Evidence GPS & Timestamp Tamper Detection	REMEDIATED	Native Android mock location filter & EXIF verification
SEC-08	Medium	Unrestricted File Upload & Large Media Denial of Service	REMEDIATED	25 MB payload ceiling, MIME allowlist (JPEG, PNG, WebP, PDF)
SEC-09	Medium	WebSocket Origin & CORS Hijacking	REMEDIATED	Strict CORS origin verification (https://networkpeer-platform.vercel.app)
SEC-10	Low	Database Direct Exposure & Shared Privileges	REMEDIATED	Private VPC subnet, SSH bastion tunnel, read-only analytics role

Detailed Remediation Actions

SEC-01: Passwordless Email OTP Rate Limiting & Storage Security

- **Vulnerability:** Vulnerability to brute-force OTP interception and email bombing.
- **Remediation:**
 - Implemented `POST /auth/email-otp/request` and `POST /auth/email-otp/verify` in `NetworkPeer-main/src/routes/auth.ts`.
 - OTPs are cryptographically hashed using SHA-256 before database storage.
 - Strict TTL of 10 minutes enforced with server-side timestamps.

- Per-email rate limit of 60 seconds cooldown between successive OTP requests.
- Lockout policy: Maximum 5 incorrect verification attempts per session, after which the OTP is permanently invalidated.

SEC-02: Server-Side Worker Review Queue Gating (Release Blocker)

- **Vulnerability:** In previous revisions, any authenticated worker could invoke `GET /jobs/:id/review-queue` and approve/reject submissions without administrative qualification.
- **Remediation:**
 - `WorkerProfile.eligibleRoles` defaults to `["collectionist"]` upon registration.
 - Collectionist gig work is **100% ungated** (workers can claim and submit data collection tasks immediately).
 - The correctionist review queue endpoints:
 - `GET /jobs/:id/review-queue`
 - `POST /submissions/:id/review` strictly verify that `workerProfile.eligible_roles` contains `'correctionist'`.
 - Non-approved workers receive **HTTP 403 Forbidden** with `ROLE_NOT_AUTHORIZED`.
 - In Android UI, the `Correct (Review)` tab in `WorkerJobPreviewScreen` is hidden unless `profile.workerProfile.eligibleRoles.contains("correctionist")`.
 - Admin endpoint `POST /admin/workers/:id/roles` provides explicit grant/revoke capabilities for authorized administrators.

SEC-03: Proprietary Model Name scrubbing & Client Leak Prevention

- **Vulnerability:** Internal LLM architecture and model identifiers (`Qwen` , `3-8B`) were visible in mobile dialogs, badges, and web client bundles.
- **Remediation:**
 - Complete elimination of model branding from all user-facing surfaces. Replaced with generic terminology: "OCR Text Extraction", "Devanagari & English Transcript", "Confidence: 98%".
 - Created automated CI script `scripts/check-model-leak.sh` which scans `apps/web/src` and `apps/android/app/src`.
 - CI verification test exits with code 0 and 0 matching strings.

SEC-04: Mandatory Full Name & Mobile Number Identity Verification

- **Vulnerability:** Users could register with blank names, leading to 404/null pointer crashes in mobile profile screens and incomplete worker accountability.
- **Remediation:**
 - Both web (`auth.verify.tsx` , `client.profile.tsx`) and mobile (`NetworkPeerApp.kt`) enforce mandatory fields:
 - `Full Name` : Required, minimum 2 characters.
 - `Mobile Number` : Required, validated for standard 10-digit / E.164 phone formats, labeled `(unverified)`.
 - Form submission is blocked if either field is missing or invalid.

- Mobile profile screen catches 404 gracefully with stored session fallback to prevent blank dash – states.

SEC-05: Web Hydration Outage & Root Error Boundary Containment

- **Vulnerability:** Hydration failure in `apps/web/src/routes/client.tsx` crashed the entire web platform.
- **Remediation:**
 - Restored complete `ClientLayout` with `PortalShell` and `<Outlet />`.
 - Guarded `useOfflineEvidenceSync()` and `RealtimeSyncBridge` with client-only mounting checks (`typeof window !== "undefined"`).
 - Ensured root error boundary isolates component crashes without tearing down full page views.

SEC-06: Escrow Funding Verification Before Job Dispatch

- **Vulnerability:** Unfunded jobs could theoretically be claimed by workers, creating unpaid labor liability.
- **Remediation:**
 - Jobs created via `POST /client/jobs` start in status `FUNDING / UNFUNDED`.
 - The worker marketplace query filters strictly for `status IN ('PUBLISHED', 'ACTIVE')` and `funding_status = 'FUNDED'`.
 - Web UI highlights payment status and routes clients directly to Stripe / Escrow funding prior to worker dispatch.

SEC-07: GPS and Timestamp Tamper Detection

- **Vulnerability:** Workers spoofing GPS coordinates or submitting recycled gallery photos.
- **Remediation:**
 - Android client checks `Location.isFromMockProvider()` on Android API 18+ and flags mock locations.
 - Camera capture enforces live timestamping and hardware EXIF location stamping.
 - Submissions are geofenced against the client's declared job radius.

SEC-08: Large Media Denial of Service & MIME Type Restriction

- **Vulnerability:** Unchecked upload sizes could exhaust storage or crash mobile memory.
- **Remediation:**
 - Strict 25 MB payload ceiling enforced both client-side and via S3 presigned POST policies.
 - Allowed MIME types strictly limited to: `image/jpeg`, `image/png`, `image/webp`, `application/pdf`, and supported short audio/video recordings.

SEC-09: CORS & WebSocket Origin Verification

- **Vulnerability:** Cross-origin WebSocket hijacking.
- **Remediation:**

- Distributed WebSocket gateway checks `Origin` header against allowlisted domains (`https://networkpeer-platform.vercel.app`).
- Strict HTTP CORS headers restrict API access to verified domains.

SEC-10: Database Access Isolation & Read-Only Least Privilege

- **Vulnerability:** Risk of accidental schema alteration or deletion from developer/analyst workstations.
 - **Remediation:**
 - AWS RDS PostgreSQL cluster is completely isolated inside private VPC subnets.
 - External access is mediated exclusively via SSH bastion local port forwarding (port 5433).
 - Dedicated read-only role `networkpeer_readonly` created with `SELECT` privileges only, preventing any `INSERT` , `UPDATE` , `DELETE` , or `DROP` commands.
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Verification & Sign-Off

The undersigned engineering lead certifies that all 10 security findings from Revision 5 have been implemented, tested, and passed: - Web typecheck: `npm --prefix apps/web run typecheck` -> **PASS (0 errors)** - Model leak CI guard: `./scripts/check-model-leak.sh` -> **PASS (0 leaks)** - Android compilation: `./gradlew assembleDevelopmentDebug` -> **BUILD SUCCESSFUL** - Device installation: Samsung Galaxy S9 (`SM-G960F`) -> **SUCCESS**